Improving Chess Reinforcement Learning Models Interpretability & Performance

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Relevant Work

- Current chess reinforcement models often suffer from lack of **interpretability** and **performance** concerns compared to analytical chess methods.
- We explore two different approaches to create more **interpretable** and **higher performing** chess Deep Learning models.

Classical Stockfish

- Utilizes the minimax algorithm with alpha-beta pruning along with an evaluation function
- Very efficient, explainable due to the deterministic function
- Doesn't quite match the reasoning and long-term planning of top level GMs

Self-Play RL Methods

AlphaZero

- AlphaZero is the poster child of chess reinforcement learning models, developed by Google DeepMind[3].
- Capable of learning any two player, turn-based game through self-play by minimizing the following objective function[2]:

$$l = (z - v)^2 - \pi^T log(p) + c||\theta||^2$$

- Provides the highest level of chess performance, but with extremely high computation cost
- Lacks explainability (why moves are chosen) due to complex NN structure for evaluating positions
- There isn't clear insight with respect to internal feature weights, what happens "under the hood"

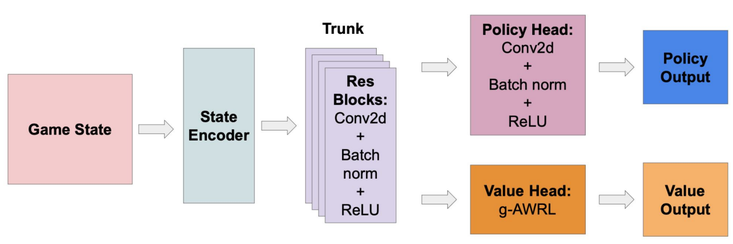
"Off The g-AWRL"

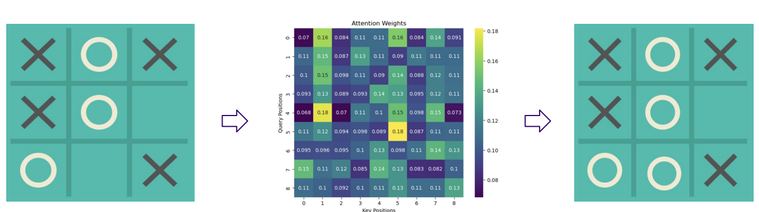
- **Generalized Attentioned Weight Reinforcement Learning** integrates **attention mechanisms** into **weighted reinforcement learning**[1].
- It weights **features *i* & *j*** in **relation** with one another as opposed to exclusively their **individual importance**:

$$v(S) = \sum_{i=1}^{L} W^i (\sum_{j=1}^{L} q(i)k(i)^T e(S_j, :))$$

Architecture & Tic Tac Toe Example

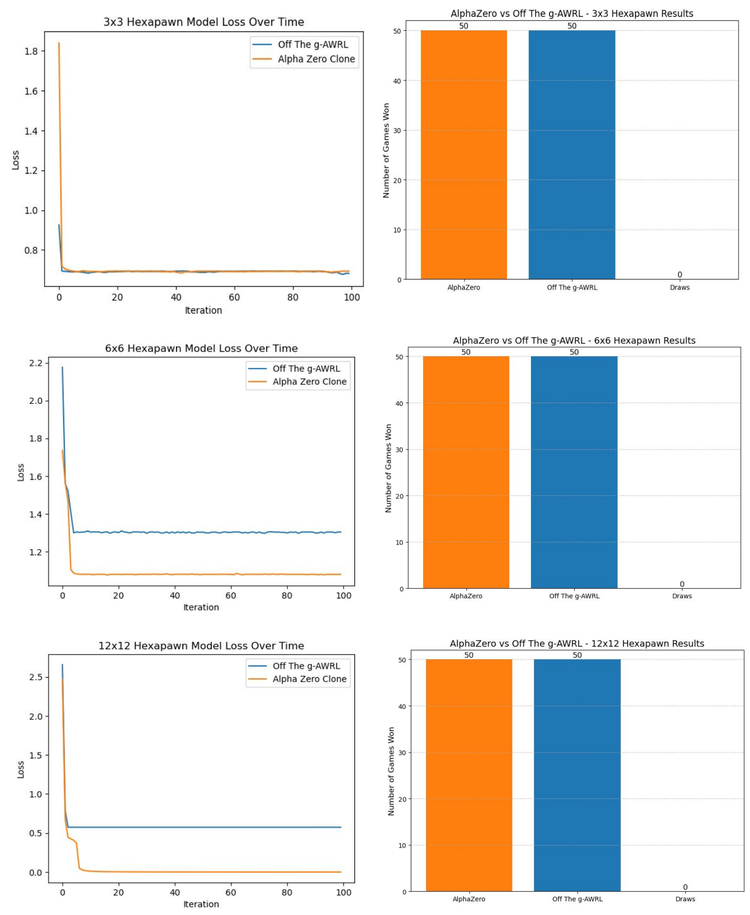
- **Off The g-AWRL** utilizes a similar architecture to AlphaZero, using g-AWRL for the value head.





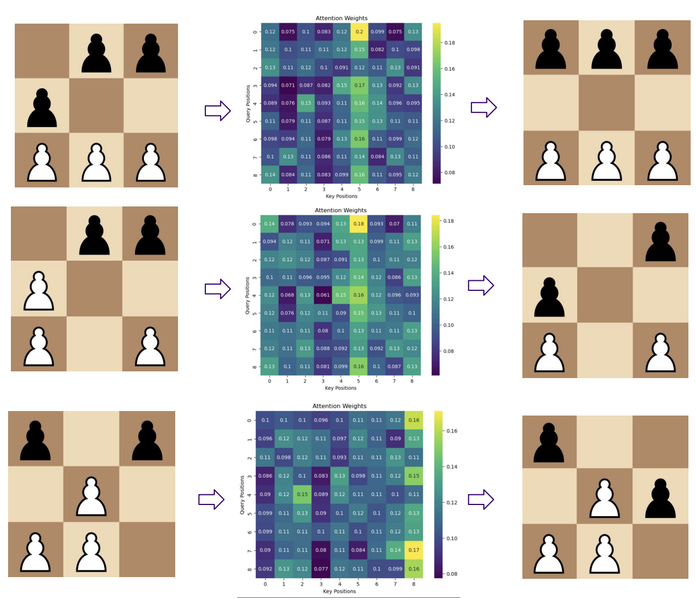
Hexapawn Evaluation

- We evaluated our **Off The g-AWRL** model against an AlphaZero model clone on **Hexapawn**, a scaled down version of chess.
- Both models quickly solved the games, **usually converging** within 10 iterations:



- Training takes **slightly longer** (**~1.25x** more than the AlphaZero clone).

g-AWRL Weight Visualizations



Efficiency Evaluation

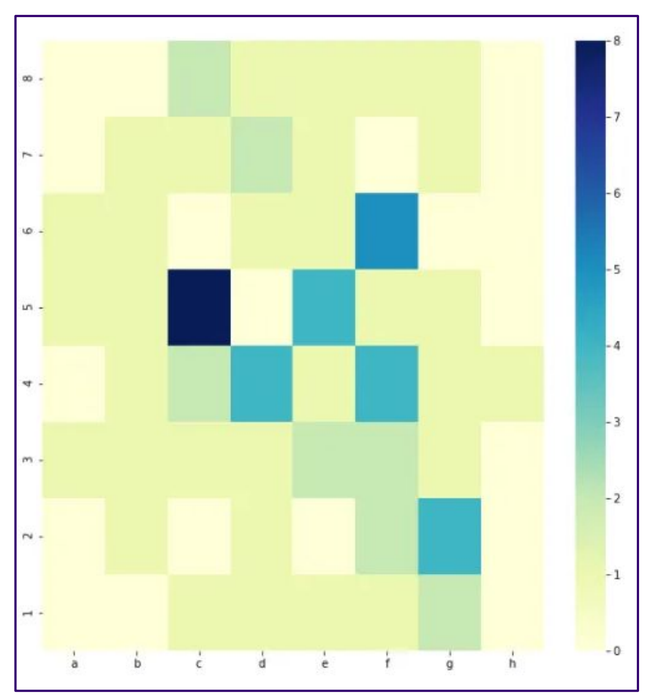
Average Move Time (seconds)

Hexapawn Game Type	AlphaZero	Off The g-AWRL
3x3	0.001395	0.001536
6x6	0.001739	0.001992
12x12	0.002120	0.002388

Data-Based RL Methods

Portable Game Notation Predictor

- **Portable Game Notation** is a method for recording chess games in plaintext format.
- We trained a language model on the **lichess** PGN dataset[4].
- We used attention across a board and across move representations to explain model decisions
- The model took about 50% more time to converge to a suboptimal solution



- We used attention across a board and across move representations to explain model decisions
- These heatmaps can help train human chess players when playing speed chess.

Discussion & Future Work

- **g-AWRL** can make RL models more **interpretable**, while giving comparable results.
- Train a captioning model that can explain its decisions in natural language using attention maps
- Train a model for question-answering tasks with chess
- Train an explainable model for chess variants (e.g. Xianqi)
- Train a model that can improve based on human (or text feedback)
- We believe that models that can explain their decisions in natural language is the best way to develop and leverage AGI.

References

[1] Lennart Bramlage and Aurelio Cortese. Generalized attention-weighted reinforcement learning. Neural Networks, 145:10–21, 2022

[2] Dominik Klein. Neural networks for chess, 2022.

[3] David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dharshan Kumaran, Thore Graepel, Timothy Lillicrap, Karen Simonyan, and Demis Hassabis. Mastering chess and shogi by self-play with a general reinforcement learning algorithm, 2017.

[4] Mitchell Jordinger. Chess game dataset (lichess) v1, 2017. This is a set of just over 20,000 games collected from a selection of users on the site Lichess.org